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On the conjecture about the nonexistence of homogeneous rotation symmetric bent functions

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Abstract
Bent functions are maximally nonlinear Boolean functions, which have wide applications due to their nice combinatorial properties. It was conjectured that there are no homogeneous rotation symmetric bent functions with degree larger than two. By investigating the connections of the short algebraic normal form of a rotation symmetric Boolean function and its derivative and nonlinearity, we obtain some new nonexistence results of homogeneous rotation symmetric bent functions, which solve the above conjecture partially.

Keywords Bent function · Rotation symmetric Boolean function · Nonlinearity · Algebraic degree · Short algebraic normal form

1 Introduction

Boolean functions play an important role in the design of many symmetric cryptosystems. In 1976, Rothaus [22] introduced a class of Boolean functions with optimal nonlinearity, which are called bent functions. These functions have important applications in cryptography, sequences, coding theory, graph theory and designs as they have nice combinatorial properties. Important surveys on bent functions related to their constructions and properties can be found in [1, 4, 6, 19].

Rotation symmetric Boolean function from a subclass of Boolean functions which are invariant under the action of the cyclic group. There is a bijective correspondence between idempotents and rotation symmetric Boolean functions by choosing a normal basis over a finite field. Rotation symmetric Boolean functions have and need less storage space, which makes the calculation of their value more efficient than for general Boolean functions, there-

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fore a search for Boolean functions with desirable cryptographic properties can be carried out quickly. For these reasons, rotation symmetric Boolean functions with nice cryptographic properties have been investigated extensively [7, 10–12, 14, 15, 20, 21, 27, 29, 32]. In particular, rotation symmetric bent functions are also well studied [2, 3, 9, 13, 17, 23, 26, 30, 34]. In [13], Gao et al. obtained a class of cubic rotation symmetric bent functions for the first time. Later, Su et al. [26] obtained n-variable rotation symmetric bent functions of any algebraic degree between 2 and $n/2$. They also obtained a class of bent idempotents with maximal algebraic degree. Zhang et al. [34] constructed some more rotation symmetric bent functions with maximal algebraic degree. Tang et al. [30] got some classes of bent idempotents over $\mathbb{F}_{2^n}$ of any algebraic degree between 2 and $n/2$. Recently, Li et al. [17] presented another class of bent idempotents with any possible algebraic degree. Early in 2004, Xia et al. [31] showed that there were no n-variable homogeneous bent functions of degree $n/2$. It is natural to study what kind of homogeneous rotation symmetric bent functions exist.

In 2008, Stănică et al. [24] studied rotation symmetric bent functions with a small number of variables and made the following conjecture according to the experimental results.

Conjecture *There are no homogeneous rotation symmetric bent functions of degree larger than 2.*

Afterwards, how to solve this conjecture received a lot of attention. In [25], Stănică et al. showed that homogeneous rotation symmetric functions generated by one orbit or some two orbits were not bent functions. They also presented a lower bound for the algebraic degree of a homogeneous rotation symmetric bent function by studying the largest distance between two consecutive indices of its terms. By investigating the relationship between the Walsh spectra of a Boolean function at partial points and the Walsh spectra of its subfunctions, Meng et al. [18] got a new lower bound of degree for homogeneous rotation symmetric bent functions. In [33], Zhang and Gao presented some new nonexistence results of homogeneous rotation symmetric bent functions by analyzing the length of consecutive indices of their terms. In [8], Cusick et al. investigated the properties of Hadamard difference sets related to rotation symmetric bent functions, and proved most cases of the conjecture when the number of variables is $2p$ (p is an odd prime). In general, how to solve this conjecture completely is an extremely difficult problem as there are few methods for establishing an effective connection between the properties of bent functions and the function's homogeneity, although bent functions have various equivalent characterizations.

In this paper, we study the conjecture from different viewpoints and obtain some new nonexistence results for the homogeneous rotation symmetric bent functions. Firstly, we show that rotation symmetric bent functions have specific values on some short orbits when the factorization of n satisfies some conditions. Secondly, we present some nonexistence results of homogeneous rotation symmetric bent functions by studying the relation between a rotation symmetric function's nonlinearity and the distance of indices of its terms. At last, we obtain some nonexistence results of cubic homogeneous rotation symmetric bent functions by investigating the balancedness of their derivatives. Our results provide new perspectives and approaches for solving the conjecture.

The rest of this paper is organized as follows. Section 2 reviews some basic notions of Boolean functions. Our main results are presented in Sect. 3. The paper is concluded in Sect. 4.

2 Preliminaries

Let $\mathbb{F}_2$ be the binary field and $\mathbb{F}_2^n$ be the n-dimensional vector space over $\mathbb{F}_2$. Let $\alpha, \beta \in \mathbb{F}_2^n$ with $\alpha = (a_0, a_1, \ldots, a_{n-1})$ and $\beta = (b_0, b_1, \ldots, b_{n-1})$. We define the support of α as $supp(\alpha) = \{0 \le i \le n-1 : a_i = 1\}$ and the Hamming weight of α as $wt(\alpha) = |supp(\alpha)|$. The complement of α is denoted by $\bar{\alpha} = (1 + a_0, 1 + a_1, \ldots, 1 + a_{n-1})$. The inner product of α and β is defined as $\alpha \cdot \beta = \sum_{i=0}^{n-1} a_i b_i$. For simplicity, denote by 0^n the all-zero vector and 1^n the all-one vector.

An n-variable Boolean function f is a mapping from $\mathbb{F}_2^n$ to $\mathbb{F}_2$. The algebraic normal form (ANF) of f is

$$f(x_0, x_1, \ldots, x_{n-1}) = \sum_{I \subseteq \{0,1,2,\ldots,n-1\}} a_I \prod_{i \in I} x_i, a_I \in \mathbb{F}_2.$$

For a set $S \subseteq \{0, 1, 2, \ldots, n-1\}$, if f contains a term $x_{i_0} x_{i_1} \cdots x_{i_{d-1}}$ with $i_0, i_1, \ldots, i_{d-1} \in S$, we say S matches f. The algebraic degree of f is defined as

$$deg(f) = \max_{I \subseteq \{0,1,2,\ldots,n-1\}} \{|I| : a_I \neq 0\}.$$

The Hamming weight of f is $wt(f) = |\{\alpha \in \mathbb{F}_2^n : f(\alpha) = 1\}|$. If $wt(f) = 2^{n-1}$, f is said to be balanced. The set of all n-variable Boolean functions is denoted by $\mathfrak{B}_n$.

Let $f \in \mathfrak{B}_n$, its Walsh transform at $\alpha \in \mathbb{F}_2^n$ is

$$W_f(\alpha) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + \alpha \cdot x}.$$

Note that f is a balanced function if $W_f(0^n) = 0$. The nonlinearity of f is

$$nl(f) = 2^{n-1} - \frac{1}{2} \max_{\alpha \in \mathbb{F}_2^n} |W_f(\alpha)|.$$

If $nl(f)$ achieves its maximal value $2^{n-1} - 2^{n/2-1}$, we call f a bent function. The derivative of f at α is

$$\Delta_\alpha f(x) = f(x + \alpha) + f(x).$$

The autocorrelation function of f at α is

$$C_f(\alpha) = \sum_{x \in \mathbb{F}_2^n} (-1)^{\Delta_\alpha f(x)}.$$

Note that f is a bent function if and only if $C_f(\alpha) = 0$ for all $\alpha \in \mathbb{F}_2^n \backslash \{0^n\}$.

For $\alpha = (a_0, a_1, \ldots, a_{n-1}) \in \mathbb{F}_2^n$ and $0 \le l \le n-1$, define

$$\rho_n^l(\alpha) = (a_l, a_{(1+l) \bmod n}, \ldots, a_{(n-1+l) \bmod n}).$$

Definition 1 Let $f \in \mathfrak{B}_n$, if $f(\rho_n^l(\alpha)) = f(\alpha)$ holds for all $\alpha \in \mathbb{F}_2^n$ and $0 \le l \le n-1$, then f is called a rotation symmetric Boolean function (RSBF).

Let $G_n(\alpha) = \{\rho_n^l(\alpha) : 0 \le l \le n-1\}$, that is, the orbit generated by α under the action of the cyclic group. An orbit is completely determined by its representative element, which is the lexicographically first element belonging to the orbit. If $|G_n(\alpha)| = n$, we call $G_n(\alpha)$ a long orbit, and if $|G_n(\alpha)| < n$, we call it a short orbit. Let g_n be the total number of orbits in $\mathbb{F}_2^n$, and h_n be the total number of long orbits in $\mathbb{F}_2^n$. Note that there are 2^{g_n} n-variable RSBFs. The following enumeration formulas of g_n and h_n are well known.

Lemma 1 [6, 16, 24] *Let $n = p_1^{n_1} p_2^{n_2} p_3^{n_3} \cdots p_t^{n_t}$ be the prime power factorization of n. Then we have*

$$g_n = \frac{1}{n} \sum_{d|n} \phi(d) 2^{\frac{n}{d}},$$

$$h_n = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d},$$

where ϕ is Euler's phi-function and μ is the Möbius function.

For $\alpha \in \mathbb{F}_2^n$, if $|G_n(\alpha)| = m$, then α can be represented as

$$\alpha = (\underbrace{\beta, \beta, \ldots, \beta}_{n/m}),$$

where β is a long orbit over $\mathbb{F}_2^m$. For simplicity, we denote α by $\alpha = \beta^{n/m}$. Let R_n be the set of representative elements of all orbits in $\mathbb{F}_2^n$, and $R_{n,m} = \{\alpha \in R_n : |G_n(\alpha)| = m\}$, that is, the set of representative elements of all orbits with size m. Note that every orbit in $\mathbb{F}_2^n$ with size m corresponds to a long orbit in $\mathbb{F}_2^m$, it follows that $|R_{n,m}| = |R_{m,m}|$. For simplicity, we set $h_m = |R_{m,m}|$. According to whether a vector and its complement are in the same orbit or not, $R_{n,m}$ can be divided into two parts $R_{n,m}^{(1)}$ and $R_{n,m}^{(2)}$:

$$R_{n,m}^{(1)} = \{\alpha \in R_{n,m} : \bar{\alpha} \notin G_n(\alpha)\}, \quad R_{n,m}^{(2)} = \{\alpha \in R_{n,m} : \bar{\alpha} \in G_n(\alpha)\}.$$

If $\alpha \in R_{n,m}^{(2)}$, then the representative element of $G_n(\bar{\alpha})$ is also in $R_{n,m}^{(2)}$. Moreover, $R_{n,m}^{(1)}$ can be further partitioned into two symmetric sets $H_{n,m}$ and $\bar{H}_{n,m}$, that is, $\alpha \in H_{n,m}$ if and only if the representative element of $G_n(\bar{\alpha})$ is in $\bar{H}_{n,m}$. Hence, $H_{n,m} \cap \bar{H}_{n,m} = \varnothing$ and $|H_{n,m}| = |\bar{H}_{n,m}|$. Let $h_m^{(1)} = |R_{n,m}^{(1)}|$ and $h_m^{(2)} = |R_{n,m}^{(2)}|$. We have the following results on $h_m^{(2)}$.

Lemma 2 [28] *Let $m = 2^t l$ be an even integer, where l is an odd integer. Then*

$$h_m^{(2)} = \begin{cases} \dfrac{2^{2^{t-1}}}{2^t}, & l = 1, \text{ i.e. } m = 2^t, \\ \dfrac{2^{m/2} - 2^{2^{t-1}}}{m}, & l \text{ is an odd prime}, \\ \dfrac{2^{m/2} - \sum\limits_{d|l, 1 \le d < l} 2^t d \cdot h_{2^t d}^{(2)}}{m}, & \text{otherwise}. \end{cases}$$

We extend the definition of ρ to the monomial $x_0 x_{i_1} \cdots x_{i_{d-1}}$ as

$$\rho_n^l(x_0 x_{i_1} \cdots x_{i_{d-1}}) = x_l x_{(i_1+l) \bmod n} \cdots x_{(i_{d-1}+l) \bmod n},$$

where $0 \le l \le n-1$. Similarly, let

$$G_n(x_0 x_{i_1} \cdots x_{i_{d-1}}) = \{\rho_n^l(x_0 x_{i_1} \cdots x_{i_{d-1}}) : 0 \le l \le n-1\}$$

be the orbit generated by the monomial $x_0 x_{i_1} \cdots x_{i_{d-1}}$. A rotation symmetric Boolean function f can be represented as its short algebraic normal form (SANF):

$$a_0 + a_1 x_0 + \sum_j a_{0j} x_0 x_j + \cdots + a_{01\ldots(n-1)} x_0 x_1 \cdots x_{n-1},$$

where $a_0, a_1, a_{0j}, \ldots, a_{01\cdots(n-1)} \in \mathbb{F}_2$, and the existence of $x_0x_{i_1}\cdots x_{i_{d-1}}$ implies that all the terms in $G_n(x_0x_{i_1}\cdots x_{i_{d-1}})$ appear in the ANF of f. In the rest of this paper, we use $[0, i_1, \ldots, i_{d-1}]$ instead of $x_0x_{i_1}\cdots x_{i_{d-1}}$ to represent the SANF of an n-variable RSBF to avoid any confusion. If f has SANF $[0, i_1, \ldots, i_{d-1}]$, we call f a monomial rotation symmetric function.

Let f be an n-variable homogeneous RSBF with degree d and denote it by

$$f = T_1 + T_2 + \cdots + T_k, \tag{1}$$

where $T_i = [0, t_1^{(i)}, t_2^{(i)}, \ldots, t_{d-1}^{(i)}]$ and $t_1^{(i)} < t_2^{(i)} < \ldots < t_{d-1}^{(i)}$, $1 \le i \le k$. Let

$$S_i = \{0, t_1^{(i)}, t_2^{(i)}, \ldots, t_{d-1}^{(i)}\}$$

for $1 \le i \le k$. Hence, denote by

$$S_i \bmod m = \{t_j^{(i)} \bmod m : 0 \le j \le d-1\}$$

for a positive integer m, where $t_0^{(i)} = 0$.

3 Nonexistence results of homogeneous rotation symmetric bent functions

This section studies the conjecture from three different ways and proves that some classes of homogeneous rotation symmetric functions cannot be bent functions.

3.1 Nonexistence results of homogeneous rotation symmetric bent functions based on the autocorrelation functions

Let $n = 2^{n_0}p_1^{n_1}p_2^{n_2}p_3^{n_3}\cdots p_t^{n_t}$ be the prime power factorization of even n and $v_2(n) = n_0$. Denote by

$$M_0 = \{2^{n_0'} : 0 \le n_0' \le n_0\}$$

and

$$M_i = \{2^{n_0'}p_1^{n_1'}p_2^{n_2'}p_3^{n_3'}\cdots p_i^{n_i'} : 0 \le n_j' \le n_j \ for\ 0 \le j \le i-1, 1 \le n_i' \le n_i\},$$

where $1 \le i \le t$. For $1 \le j < i \le t$ and $m \in M_j$, we have $p_j \mid m$ and $p_i \nmid m$. It follows that

$$\sum_{x\in\mathbb{F}_2^n}(-1)^{f(x)+f(x+1^n)} = \sum_{i=0}^{t}\sum_{m\in M_i} m \sum_{x\in R_{n,m}}(-1)^{f(x)+f(x+1^n)}.$$

Theorem 1 *Let n be even and $n = 2^{n_0}p_1^{n_1}p_2^{n_2}p_3^{n_3}\cdots p_t^{n_t}$ be its prime power factorization with $p_1 > 2^{2^{n_0}}$ and $p_i > 2^{2^{n_0}p_1^{n_1}\cdots p_{i-1}^{n_{i-1}}}$ for any $2 \le i \le t$. Let f be an n-variable rotation symmetric bent function. Then the following results hold.*

1. *If $n_0 = 1$, then $f(0^n) \ne f(1^n)$.*
2. *If $n_0 = 2$, then $f(0^n) = f(1^n)$ and $f((1,0,0,0)^{n/4}) \ne f((1,1,1,0)^{n/4})$.*
2. *If $n_0 \ge 3$, then $f(0^n) = f(1^n)$ and $f((1,1,1,0)^{n/4}) = f((1,0,0,0)^{n/4})$.*

Proof As f is bent, then

$$\begin{aligned}\sum_{x\in\mathbb{F}_2^n}(-1)^{f(x)+f(x+1^n)} &= \sum_{i=0}^{t}\sum_{m\in M_i} m \sum_{x\in R_{n,m}}(-1)^{f(x)+f(x+1^n)}\\ &= \sum_{i=0}^{t-1}\sum_{m\in M_i} m \sum_{x\in R_{n,m}}(-1)^{f(x)+f(x+1^n)}\\ &\quad + \sum_{m\in M_t} m \sum_{x\in R_{n,m}}(-1)^{f(x)+f(x+1^n)}\\ &= 0.\end{aligned}$$

Obviously, p_t divides $\sum_{m\in M_t} m \sum_{x\in R_{n,m}}(-1)^{f(x)+f(x+1^n)}$. Since

$$p_t > 2^{2^{n_0}p_1^{n_1}\cdots p_{t-1}^{n_{t-1}}} = \sum_{i=0}^{t-1}\sum_{m\in M_i} m|R_{n,m}|,$$

we have

$$\sum_{m\in M_t} m \sum_{x\in R_{n,m}}(-1)^{f(x)+f(x+1^n)} = 0$$

and

$$\sum_{i=0}^{t-1}\sum_{m\in M_i} m \sum_{x\in R_{n,m}}(-1)^{f(x)+f(x+1^n)} = 0.$$

Hence, since $p_i > 2^{2^{n_0}p_1^{n_1}\cdots p_{i-1}^{n_{i-1}}} = \sum_{j=0}^{i-1}\sum_{m\in M_j} m|R_{n,m}|$ for $i \geq 1$, we can deduce that

$$\sum_{i=0}^{n_0} 2^i \sum_{x\in R_{n,2^i}}(-1)^{f(x)+f(x+1^n)} = 0$$

by considering $p_{t-1}, p_{t-2}, \ldots, p_1$ similarly as p_t.

If $n_0 = 1$, we have

$$2(-1)^{f(0)+f(1^n)} + 2(-1)^{f((0,1)^{n/2})+f((1,0)^{n/2})} = 0.$$

Since $f((0,1)^{n/2}) = f((1,0)^{n/2})$, then $f(0^n) + f(1^n) = 1$.

If $n_0 = 2$, we have

$$\begin{aligned}\sum_{i=0}^{2} 2^i \sum_{x\in R_{n,2^i}}(-1)^{f(x)+f(x+1^n)} &= 2(-1)^{f(0^n)+f(1^n)} + 2(-1)^{f((0,1)^{n/2})+f((1,0)^{n/2})}\\ &\quad + 4(-1)^{f((1,1,0,0)^{n/4})+f((0,0,1,1)^{n/4})}\\ &\quad + 8(-1)^{f((0,0,0,1)^{n/4})+f((1,1,1,0)^{n/4})}\\ &= 6 + 2(-1)^{f(0^n)+f(1^n)} + 8(-1)^{f((1,0,0,0)^{n/4})+f((1,1,1,0)^{n/4})}\\ &= 0.\end{aligned}$$

It follows that $f(0^n) + f(1^n) = 0$ and $f((1,0,0,0)^{n/4}) + f((1,1,1,0)^{n/4}) = 1$.

Note that $h_8^{(2)} = 2$. If $n_0 = 3$, we have

$$\begin{aligned}\sum_{i=0}^{n_0} 2^i \sum_{x\in R_{n,2^i}} (-1)^{f(x)+f(x+1^n)} &= 2(-1)^{f(0^n)+f(1^n)} + 2(-1)^{f((0,1)^{n/2})+f((1,0)^{n/2})} \\ &\quad + 4(-1)^{f((1,1,0,0)^{n/4})+f((0,0,11)^{n/4})} \\ &\quad + 8(-1)^{f((0,0,0,1)^{n/4})+f((1,1,1,0)^{n/4})} \\ &\quad + 16 \sum_{x\in H_{n,8}^{(1)}} (-1)^{f(x)+f(x+1^n)} \\ &\quad + 8 \sum_{x\in R_{n,8}^{(2)}} (-1)^{f(x)+f(x+1^n)} \\ &= 22 + 2(-1)^{f(0^n)+f(1^n)} + 8(-1)^{f((1,0,0,0)^{n/4})+f((1,1,1,0)^{n/4})} \\ &\quad + 16 \sum_{x\in H_{n,8}^{(1)}} (-1)^{f(x)+f(x+1^n)} \\ &= 0.\end{aligned}$$

One can verify that $H_{n,8}^{(1)}$ is even, then 32 divides $16 \sum_{x\in H_{n,8}^{(1)}} (-1)^{f(x)+f(x+1^n)}$. It follows that

$$f(0^n) + f(1^n) = f((1,0,0,0)^{n/4}) + f((1,1,1,0)^{n/4}) = 0.$$

Note that $h_m^{(2)} = 16$. If $n_0 \geq 4$, we have

$$\begin{aligned}&\sum_{i=0}^{n_0} 2^i \sum_{x\in R_{n,2^i}} (-1)^{f(x)+f(x+1^n)} \\ &= 22 + 2(-1)^{f(0^n)+f(1^n)} + 8(-1)^{f((0,0,0,1)^{n/4})+f((1,1,1,0)^{n/4})} \\ &\quad + 16 \sum_{x\in H_{n,8}^{(1)}} (-1)^{f(x)+f(x+1^n)} + 32 \sum_{x\in H_{n,16}^{(1)}} (-1)^{f(x)+f(x+1^n)} \\ &\quad + 16 \sum_{x\in R_{n,16}^{(2)}} (-1)^{f(x)+f(x+1^n)} + \sum_{i=5}^{n_0} 2^i \sum_{x\in R_{n,2^i}} (-1)^{f(x)+f(x+1^n)} \\ &= 278 + 2(-1)^{f(0^n)+f(1^n)} + 8(-1)^{f((1,0,0,0)^{n/4})+f((1,1,1,0)^{n/4})} \\ &\quad + 16 \sum_{x\in H_{n,8}^{(1)}} (-1)^{f(x)+f(x+1^n)} + 32 \sum_{x\in H_{n,16}^{(1)}} (-1)^{f(x)+f(x+1^n)} \\ &\quad + \sum_{i=5}^{n_0} 2^i \sum_{x\in R_{n,2^i}} (-1)^{f(x)+f(x+1^n)} \\ &= 0.\end{aligned}$$

Since $H_{n,8}^{(1)}$ is even, then 32 divides $16\sum_{x\in H_{n,8}^{(1)}}(-1)^{f(x)+f(x+1^n)}$, it follows that 32 divides $278+2(-1)^{f(0^n)+f(1^n)}+8(-1)^{f((1,0,0,0)^{n/4})+f((1,1,1,0)^{n/4})}$. Thus, we have

$$f(0^n)+f(1^n)=f((1,0,0,0)^{n/4})+f((1,1,1,0)^{n/4})=0.$$

This completes the proof. □

Lemma 3 *Let $f=[0,r_1,r_2,\ldots,r_{d-1}]$ be an n-variable monomial RSBF with degree d and $S=\{r_i \bmod 4 : 0\le i\le d-1\}$, where $r_0=0$. We have the following results.*

1. *$f(0^n)\neq f(1^n)$ if and only if $|G_n(x_0x_{r_1}x_{r_2}\cdots x_{r_{d-1}})|$ is odd.*
2. *Assume that $v_2(n)\ge 2$. $f((1,0,0,0)^{n/4})\neq f((1,1,1,0)^{n/4})$ if and only if one of the following conditions is satisfied.*

(1). $|G_n(x_0x_{r_1}x_{r_2}\cdots x_{r_{d-1}})|\equiv 4 \bmod 8$ *and* $|S|=3$.
(2). $|G_n(x_0x_{r_1}x_{r_2}\cdots x_{r_{d-1}})|\equiv 2 \bmod 4$ *and* $|S|=2$.

Proof 1. Note that $f(0^n)=0$. Hence, $f(1^n)=1$ if $G_n(x_0x_{r_1}x_{r_2}\cdots x_{r_{d-1}})$ has an odd number of terms, then the result follows.

2. Let $L=|G_n(x_0x_{r_1}x_{r_2}\cdots x_{r_{d-1}})|$ and represent f as $f(x)=\sum_{i=0}^{L-1}h_i(x)$, where $h_i(x)=\rho_n^i(x_0x_{r_1}x_{r_2}\cdots x_{r_{d-1}})$. First note that

$$kL \bmod n,(r_j+kL) \bmod n\in\{0,r_1,r_2,\ldots,r_{d-1}\}$$

for any $k\ge 0$ and $1\le j\le d-1$. For $0\le i\le L-1$, it is clear that $h_i((1,0,0,0)^{n/4})=1$ if and only if

$$\{i,(i+r_1) \bmod n,\ldots,(i+r_{d-1}) \bmod n\}\subseteq\{0,4,8,12,\ldots,4n-4\},$$

and $h_i((1,1,1,0)^{n/4})=1$ if and only if

$$\{i,(i+r_1) \bmod n,\ldots,(i+r_{d-1}) \bmod n\}\cap\{3,7,11,15,\ldots,4n-3\}=\varnothing.$$

The discussion is divided into three cases.

Case 1: L is odd. Let $0\le i\le L-1$. On one hand, since i and $(i+L) \bmod n$ have different parity, then

$$\{i,(i+r_1) \bmod n,\ldots,(i+r_{d-1}) \bmod n\}\nsubseteq\{0,4,8,12,\ldots,4n-4\}.$$

It follows that $h_i((1,0,0,0)^{n/4})=0$. Hence, we have

$$f((1,0,0,0)^{n/4})=\sum_{i=0}^{L-1}h_i((1,0,0,0)^{n/4})=0.$$

On the other hand, we have

$$\{i,(i+r_1) \bmod n,\ldots,(i+r_{d-1}) \bmod n\}\cap\{3,7,11,15,\ldots,4n-3\}\neq\varnothing,$$

it follows that $h_i((1,1,1,0)^{n/4})=0$. Hence, we have

$$f((1,1,1,0)^{n/4})=\sum_{i=0}^{L-1}h_i((1,1,1,0)^{n/4})=0.$$

Case 2: $L\equiv 0 \bmod 4$. For $0\le i\le L-5$, we have

$$h_i((1,0,0,0)^{n/4})=h_{i+4}((1,0,0,0)^{n/4}),\ h_i((1,1,1,0)^{n/4})=h_{i+4}((1,1,1,0)^{n/4}).$$

If $L \equiv 0 \bmod 8$, then

$$f((1,0,0,0)^{n/4}) = \frac{n}{4}\sum_{i=0}^{3} h_i((1,0,0,0)^{n/4}) = 0,$$
$$f((1,1,1,0)^{n/4}) = \frac{n}{4}\sum_{i=0}^{3} h_i((1,1,1,0)^{n/4}) = 0.$$

If $L \equiv 4 \bmod 8$, then

$$f((1,0,0,0)^{n/4}) = \sum_{i=0}^{3} h_i((1,0,0,0)^{n/4}),\ f((1,1,1,0)^{n/4}) = \sum_{i=0}^{3} h_i((1,1,1,0)^{n/4}).$$

When $|S| = 1$, i.e., $r_i \bmod 4 \equiv 0$ for any $1 \le i \le d-1$, we have

$$\sum_{i=0}^{3} h_i((1,0,0,0)^{n/4}) = \sum_{i=0}^{3} h_i((1,1,1,0)^{n/4}) = 1.$$

It follows that $f((1,0,0,0)^{n/4}) = f((1,1,1,0)^{n/4}) = 1$.

When $|S| = 2$, we have

$$\sum_{i=0}^{3} h_i((1,0,0,0)^{n/4}) = \sum_{i=0}^{3} h_i((1,1,1,0)^{n/4}) = 0.$$

It follows that $f((1,0,0,0)^{n/4}) = f((1,1,1,0)^{n/4}) = 0$.

When $|S| = 3$, we have

$$\sum_{i=0}^{3} h_i((1,0,0,0)^{n/4}) = 0, \sum_{i=0}^{3} h_i((1,1,1,0)^{n/4}) = 1.$$

It follows that $f((1,0,0,0)^{n/4}) \neq f((1,1,1,0)^{n/4})$.

Case 3: $L \equiv 2 \bmod 4$. For $0 \le i \le L-1$, since i and $(i+L) \bmod n$ cannot appear in the set $\{0,4,8,12,\ldots,4n-4\}$ at the same time, then $h_i((1,0,0,0)^{n/4}) = 0$. Thus, $f((1,0,0,0)^{n/4}) = 0$. Note that $\{L, 2L, \ldots, n-L\} \subseteq \{0, r_1, \ldots, r_{d-1}\}$. Hence, we have $|S| = 2$ or $|S| = 4$. If $|S| = 2$, we have $\sum_{i=0}^{3} h_i((1,1,1,0)^{n/4}) = 0$. It follows that

$$\begin{aligned} f((1,1,1,0)^{n/4}) &= \sum_{i=0}^{L-3} h_i((1,1,1,0)^{n/4}) + h_{L-2}((1,1,1,0)^{n/4}) + h_{L-1}((1,1,1,0)^{n/4}) \\ &= h_0((1,1,1,0)^{n/4}) + h_1((1,1,1,0)^{n/4}) \\ &= 1. \end{aligned}$$

If $|S| = 4$, we have $h_i((1,1,1,0)^{n/4}) = 0$ for all $0 \le i \le L-1$. It follows that $f((1,1,1,0)^{n/4}) = 0$.

This completes the proof. □

Combining Theorem 1 and Lemma 3, we have the following result.

Theorem 2 *Let $n = 2^{n_0} p_1^{n_1} p_2^{n_2} p_3^{n_3} \cdots p_t^{n_t}$ with $p_1 > 2^{2^{n_0}}$ and $p_i > 2^{2^{n_0} p_1^{n_1} \cdots p_{i-1}^{n_{i-1}}}$ for any $2 \le i \le t$. Let f be an n-variable homogeneous RSBF in (1) with SANF $f = T_1 + T_2 + \cdots + T_k$. Denote by*

$$N_1 = |\{1 \le i \le k : |G_n(T_i)| \equiv 1 \bmod 2\}|,$$
$$N_2 = |\{1 \le i \le k : |G_n(T_i)| \equiv 4 \bmod 8 \text{ and } |S_i \bmod 4| = 3\}|,$$
$$N_3 = |\{1 \le i \le k : |G_n(T_i)| \equiv 2 \bmod 4 \text{ and } |S_i \bmod 4| = 2\}|.$$

1. *When $n_0 = 1$, i.e., $v_2(n) = 1$, if N_1 is even, f cannot be a bent function.*
2. *When $n_0 = 2$, i.e., $v_2(n) = 2$, if N_1 is odd or $N_2 + N_3$ is even, f cannot be a bent function.*
3. *When $n_0 \ge 3$, i.e., $v_2(n) \ge 3$, if N_1 is odd or $N_2 + N_3$ is odd, f cannot be a bent function.*

Example 1 1. Let $n = 50$ and f be a 50-variable homogeneous RSBF with SANF

$$f = T_1 + T_2 + T_3 + T_4,$$

where

$$T_1 = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9], T_2 = [0, 1, 2, 3, 4, 5, 6, 7, 8, 10],$$
$$T_3 = [0, 5, 10, 15, 20, 25, 30, 35, 40, 45], T_4 = [0, 1, 2, 3, 4, 25, 26, 27, 28, 29].$$

Note that $\deg(f) = 10$ and

$$|G_n(T_1)| = |G_n(T_2)| = 50, |G_n(T_3)| = 5, |G_n(T_4)| = 25.$$

It follows that $N_1 = 2$. According to item 1 in Theorem 2, f is not a bent function.

2. Let $n = 68$ and f be a 68-variable homogeneous RSBF with SANF

$$f = T_1 + T_2 + T_3 + T_4,$$

where

$$T_1 = \{0, 1, 2, 5, 9, 13, 17, 21, 25, 29\}, T_2 = \{0, 1, 2, 6, 10, 14, 18, 22, 26, 30\},$$
$$T_3 = \{0, 2, 3, 6, 10, 14, 18, 22, 26, 30\}, T_4 = \{0, 1, 3, 4, 8, 12, 16, 20, 24, 28\}.$$

Note that $\deg(f) = 10$ and

$$|G_n(T_1)| \equiv |G_n(T_2)| \equiv |G_n(T_3)| \equiv |G_n(T_4)| \equiv 4 \bmod 8.$$

Hence, we have

$$|S_1 \bmod 4| = |S_2 \bmod 4| = |S_3 \bmod 4| = |S_4 \bmod 4| = 3.$$

It follows that $N_2 = 4$ and $N_3 = 0$. According to item 2 in Theorem 2, f is not a bent function.

3. Let $n = 8 \times 257 = 2056$ and f be a 2056-variable homogeneous RSBF with SANF

$$f = T_1 + T_2 + T_3 + T_4,$$

where

$$T_1 = \{0, 1, 2, 3, 4, 5, 6, 7\}, T_2 = \{0, 1, 2, 3, 4, 5, 6, 8\},$$
$$T_3 = \{0, 1, 2, 3, 4, 5, 6, 9\}, T_4 = \{0, 257, 514, 771, 1028, 1285, 1542, 1799\}.$$

Note that $\deg(f) = 8$ and

$$|G_n(T_1)| = |G_n(T_2)| = |G_n(T_3)| = 2056, |G_n(T_4)| = 257.$$

It follows that $N_1 = 1$. According to item 3 in Theorem 2, f is not a bent function.

The following result follows from Theorem 2 immediately.

Corollary 1 *Let $n = 2p_1^{n_1}p_2^{n_2}p_3^{n_3}\cdots p_t^{n_t}$ with $p_i > 2^{2p_1^{n_1}\cdots p_{i-1}^{n_{i-1}}}$ for any $2 \le i \le t$. Let f be an n-variable homogeneous rotation symmetric function with degree d. If f is bent, then d is even.*

3.2 Nonexistence results of rotation symmetric bent functions based on the distance of indices in their terms

In this section, we present some nonexistence results of rotation symmetric bent functions by studying the relation between the rotation symmetric function's nonlinearity and the distance of indices of its terms. The following lemma is crucial to our nonexistence results of rotation symmetric bent functions.

Lemma 4 [35] *Let f be a Boolean function over $\mathbb{F}_2^n$ and $I \subseteq \{0, 1, 2, \ldots, n-1\}$ such that I does not match f. Then the nonlinearity of f satisfies $N_f \le 2^{n-1} - 2^{s-1}$, where $s = |I|$.*

We next present some necessary results. For a set $W \subseteq \{0, 1, 2, \ldots, n-1\}$ with size k, we can arrange the elements in W ascendingly as $r_1 < r_2 < \cdots < r_k$. Denote by

$$\begin{aligned} D_W &= \max\{r_2 - r_1, r_3 - r_2, \ldots, r_{k-1} - r_{k-2}, r_k - r_{k-1}\}, \\ \mathcal{D}_W &= \max\{r_2 - r_1, r_3 - r_2, \ldots, r_{k-1} - r_{k-2}, r_k - r_{k-1}, r_1 + n - r_k\}. \end{aligned} \tag{2}$$

For $f(x)$ in (1), denote by

$$M_f = \max\{D_{S_1}, D_{S_2}, \ldots, D_{S_k}\}, \tag{3}$$

$$\mathcal{M}_f = \max\{\mathcal{D}_{S_1}, \mathcal{D}_{S_2}, \ldots, \mathcal{D}_{S_k}\}. \tag{4}$$

If $\mathcal{M}_f = \mathcal{D}_{S_i}$ for some i, we say that T_i generates the largest distance of f.

Let $\mathfrak{R} = \{R_1, R_2, \ldots, R_k\}$, where $R_i = \{r_1^{(i)}, r_2^{(i)}, \ldots, r_{n_i}^{(i)}\} \subseteq S_i$, $1 \le i \le k$. Assume that $n_i \ge 2$ and $r_1^{(i)} < r_2^{(i)} < \ldots < r_{n_i}^{(i)}$ for all $1 \le i \le k$. Hence, denote by

$$\Delta_{\mathfrak{R}} = \max\{D_{R_1}, D_{R_2}, \ldots, D_{R_k}\}, \tag{5}$$

$$\Gamma_{\mathfrak{R}} = \min\{r_{n_1}^{(1)} - r_1^{(1)}, r_{n_2}^{(2)} - r_1^{(2)}, \ldots, r_{n_k}^{(k)} - r_1^{(k)}\}. \tag{6}$$

Define an n-variable RSBF deduced by $\mathfrak{R}$ as

$$\begin{aligned} g_{\mathfrak{R}} =& [0, r_2^{(1)} - r_1^{(1)}, r_3^{(1)} - r_1^{(1)}, \ldots, r_{n_1}^{(1)} - r_1^{(1)}] \\ &+ [0, r_2^{(2)} - r_1^{(2)}, r_3^{(2)} - r_1^{(2)}, \ldots, r_{n_2}^{(2)} - r_1^{(2)}] \\ &+ \cdots + [0, r_2^{(k)} - r_1^{(k)}, r_3^{(k)} - r_1^{(k)}, \ldots, r_{n_k}^{(k)} - r_1^{(k)}]. \end{aligned} \tag{7}$$

One can easily prove the following result.

Lemma 5 *Let $f(x)$ and $g_{\mathfrak{R}}(x)$ be defined in (1) and (7) respectively. For a set $W \subseteq \{0, 1, 2, \ldots, n-1\}$, if W doesn't match $g_{\mathfrak{R}}$, then W doesn't match f.*

For a given homogeneous RSBF f in (1), when we use Lemma 4 to prove it is not a bent function, the key point is to find a set I with size larger than $n/2+1$ such that I doesn't match f, that is, there are no S_i such that $S_i \subseteq I$. To construct such I, we consider the maximal distance between two consecutive indices of all terms in f. Specifically, if this maximal value doesn't appear as a gap of two elements in I, then I doesn't match f. To this end, we now define a sequence as

$$\begin{aligned}
&\underbrace{1, 2, 3, \ldots, \Gamma_{\mathfrak{R}} - 1}_{\Gamma_{\mathfrak{R}}-1}, \underbrace{\Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}}, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} + 1, \ldots, 2\Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} - 2}_{\Gamma_{\mathfrak{R}}-1},\\
&\underbrace{2\Gamma_{\mathfrak{R}} + 2\Delta_{\mathfrak{R}} - 1, 2\Gamma_{\mathfrak{R}} + 2\Delta_{\mathfrak{R}}, \ldots, 3\Gamma_{\mathfrak{R}} + 2\Delta_{\mathfrak{R}} - 3}_{\Gamma_{\mathfrak{R}}-1}, \ldots,\\
&\underbrace{t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1), t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - t + 2, \ldots, (t+1)\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t+1)}_{\Gamma_{\mathfrak{R}}-1}, \ldots
\end{aligned} \tag{8}$$

which can be seen as a contention of blocks with size $\Gamma_{\mathfrak{R}} - 1$. Let's consider the following set $I' \subseteq \{0, 1, 2, \ldots, n-1\}$ deduced by the sequence (8)

$$\begin{aligned}
I' = \{&\underbrace{1, 2, \ldots, \Gamma_{\mathfrak{R}} - 1}_{\Gamma_{\mathfrak{R}}-1}, \underbrace{\Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}}, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} + 1, \ldots, 2\Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} - 2}_{\Gamma_{\mathfrak{R}}-1}, \ldots,\\
&\underbrace{t'\Gamma_{\mathfrak{R}} + t'\Delta_{\mathfrak{R}} - (t'-1), t'\Gamma_{\mathfrak{R}} + t'\Delta_{\mathfrak{R}} - t' + 2, \ldots, t'\Gamma_{\mathfrak{R}} + t'\Delta_{\mathfrak{R}} - (t'-1) + L - 1}_{L}\},
\end{aligned} \tag{9}$$

where $1 \le L \le \min\{\Gamma_{\mathfrak{R}} - 1, n - \Delta_{\mathfrak{R}} + t' - t'\Gamma_{\mathfrak{R}} + t'\Delta_{\mathfrak{R}}\}$. The condition $L \le n - \Delta_{\mathfrak{R}} + t' - t'\Gamma_{\mathfrak{R}} + t'\Delta_{\mathfrak{R}}$ implies that

$$n + 1 - (t'\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t'-1) + L - 1) \ge \Delta_{\mathfrak{R}} + 1.$$

Note that I' can be seen as a union of t' blocks with size $\Gamma_{\mathfrak{R}} - 1$ and one block with size L.

Lemma 6 *Let f and I' be defined in (1) and (9), then I' doesn't match f.*

Proof We first show that I' doesn't match $g_{\mathfrak{R}}$ in (7). Assume that I' matches one term in $G_n(x_0 x_{r_2^{(i)} - r_1^{(i)}} x_{r_3^{(i)} - r_1^{(i)}} \cdots x_{r_{n_i}^{(i)} - r_1^{(i)}})$ for some i, that is, there exists l such that

$$\{l, (r_2^{(i)} - r_1^{(i)} + l) \bmod n, \ldots, (r_{n_i}^{(i)} - r_1^{(i)} + l) \bmod n\} \subseteq I'.$$

The discussion is divided into two cases.

Case 1: $r_{n_i}^{(i)} - r_1^{(i)} + l < n$. If the set

$$\{l, r_2^{(i)} - r_1^{(i)} + l, r_3^{(i)} - r_1^{(i)} + l, \ldots, r_{n_1}^{(i)} - r_1^{(i)} + l\}$$

belongs to one block of I', we have

$$t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1) \le l < r_{n_i}^{(i)} - r_1^{(i)} + l \le (t+1)\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t+1)$$

for some t or

$$t'\Gamma_{\mathfrak{R}} + t'\Delta_{\mathfrak{R}} - (t'-1) \le l < r_{n_i}^{(i)} - r_1^{(i)} + l \le t'\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t'-1) + L - 1,$$

which implies that $r_{n_i}^{(i)} - r_1^{(i)} \le \Gamma_{\mathfrak{R}} - 1$. This contradicts the fact that $\Gamma_{\mathfrak{R}} \le r_{n_i}^{(i)} - r_1^{(i)}$.

If the set

$$\{l, r_2^{(i)} - r_1^{(i)} + l, r_3^{(i)} - r_1^{(i)} + l, \ldots, r_{n_1}^{(i)} - r_1^{(i)} + l\}$$

belongs to two different blocks of I', then there exists t such that

$$r_t^{(i)} - r_1^{(i)} + l - (r_{t-1}^{(i)} - r_1^{(i)} + l) \geq \Delta_{\mathfrak{R}} + 1,$$

that is, $r_t^{(i)} - r_{t-1}^{(i)} \geq \Delta_{\mathfrak{R}} + 1$, which contradicts the fact that $\Delta_{\mathfrak{R}} \geq r_t^{(i)} - r_{t-1}^{(i)}$.

Case 2: $r_{n_i}^{(i)} - r_1^{(i)} + l > n$. Then there exists t such that $r_{t-1}^{(i)} - r_1^{(i)} + l < n$ and $r_t^{(i)} - r_1^{(i)} + l \geq n$. According to the fact that

$$n + 1 - (t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1) + l - 1) \geq \Delta_{\mathfrak{R}} + 1,$$

we have $r_t^{(i)} - r_{t-1}^{(i)} \geq \Delta_{\mathfrak{R}} + 1$, which contradicts the fact that $\Delta_{\mathfrak{R}} \geq r_t^{(i)} - r_{t-1}^{(i)}$.

Consequently, I' doesn't match $g_{\mathfrak{R}}$, it follows from Lemma 5 that I' doesn't match f. □

Theorem 3 *Let f, $\Delta_{\mathfrak{R}}$ and $\Gamma_{\mathfrak{R}}$ be defined in (1), (5) and (6) respectively. If one of the following conditions is satisfied, then f is not a bent function:*

1. $n - \Gamma_{\mathfrak{R}} < \Delta_{\mathfrak{R}} < n/2 - 1$;
2. *there exists $t \geq 1$ such that*
$$\frac{n + t - (t+1)\Gamma_{\mathfrak{R}}}{t+1} < \Delta_{\mathfrak{R}} < \min\{\frac{n + (t+1)(1-\Gamma_{\mathfrak{R}})}{t}, \frac{n/2 - 1}{t+1}\};$$
3. *there exists $t \geq 1$ such that*
$$\frac{n + (t+1)(1-\Gamma_{\mathfrak{R}})}{t} \leq \Delta_{\mathfrak{R}} < \min\{\frac{n + t(1-\Gamma_{\mathfrak{R}})}{t+1}, \frac{n/2 - 1}{t+1}\};$$
4. *there exists $t \geq 1$ such that $\Gamma_{\mathfrak{R}} > \frac{n/2+1}{t} + 1$ and*
$$\max\{\frac{n + (t+1)(1-\Gamma_{\mathfrak{R}})}{t}, \frac{n + (t-1) - t\Gamma_{\mathfrak{R}}}{t+1}\} \leq \Delta_{\mathfrak{R}} \leq \frac{n + (t-1) - t\Gamma_{\mathfrak{R}}}{t}.$$

Proof Let's consider the relation between n and the sequence in (9). The discussion can be divided into three cases.

Case 1: n is not in a block of I, i.e.

$$(t+1)\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t+1) < n < (t+1)\Gamma_{\mathfrak{R}} + (t+1)\Delta_{\mathfrak{R}} - t \tag{10}$$

for some $t \geq 0$. In this case, we construct a set $I \subseteq \{0, 1, 2, 3, \ldots, n-1\}$ as

$$I = \{1, 2, 3, \ldots, \Gamma_{\mathfrak{R}} - 1, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}}, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} + 1, \ldots, 2\Gamma_{\mathfrak{R}} + 2\Delta_{\mathfrak{R}} - 1, \ldots,$$
$$t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1), t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - t + 2, \ldots, n + 1 - \Delta_{\mathfrak{R}} - 1\}.$$

Note that

$$\begin{aligned} |I| &= t(\Gamma_{\mathfrak{R}} - 1) + n + 1 - \Delta_{\mathfrak{R}} - 1 - (t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1)) + 1 \\ &= n - (t+1)\Delta_{\mathfrak{R}}. \end{aligned}$$

By Lemma 6, we can deduce that I doesn't match f. According to Lemma 4, if $|I| > n/2+1$, that is, $n - (t+1)\Delta_{\mathfrak{R}} < n/2 + 1$, then f is not a bent function. Note that

$$|I| > n/2 + 1 \Leftrightarrow \Delta_{\mathfrak{R}} < \frac{n/2 - 1}{t+1}. \tag{11}$$

For the inequality (10), if $t = 0$, we have

$$\Delta_{\mathfrak{R}} > n - \Gamma_{\mathfrak{R}}; \tag{12}$$

if $t \geq 1$,

$$\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t+1) < n < (t+1)\Gamma_{\mathfrak{R}} + (t+1)\Delta_{\mathfrak{R}} - t \tag{13}$$

is equivalent to

$$\frac{n + t - (t+1)\Gamma_{\mathfrak{R}}}{t+1} < \Delta_{\mathfrak{R}} < \frac{n + (t+1)(1-\Gamma_{\mathfrak{R}})}{t}.$$

Combining (11),(12) and (13), we obtain items 1 and 2.

Case 2: n is in a block of I, i.e.,

$$t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1) \leq n \leq (t+1)\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t+1)$$

for some $t \geq 1$. Note that

$$t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1) \leq n \leq (t+1)\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t+1)$$

is equivalent to

$$\frac{n + (t+1)(1-\Gamma_{\mathfrak{R}})}{t} \leq \Delta_{\mathfrak{R}} \leq \frac{n + (t-1) - t\Gamma_{\mathfrak{R}}}{t}. \tag{14}$$

This case can be divided into two subcases.

Case 2.1: $n + 1 - \Delta_{\mathfrak{R}} > t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1)$. For this subcase, we construct a set $I \subseteq \{0, 1, 2, 3, \ldots, n-1\}$ as

$$\begin{aligned} I = \{&1, 2, 3, \ldots, \Gamma_{\mathfrak{R}} - 1, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}}, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} + 1, \ldots, 2\Gamma_{\mathfrak{R}} + 2\Delta_{\mathfrak{R}} - 1, \ldots, \\ &t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1), t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - t + 2, \ldots, n + 1 - \Delta_{\mathfrak{R}} - 1\}. \end{aligned}$$

We have

$$\begin{aligned} |I| &= t(\Gamma_{\mathfrak{R}} - 1) + n + 1 - \Delta_{\mathfrak{R}} - 1 - (t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1)) + 1 \\ &= n - (t+1)\Delta_{\mathfrak{R}}. \end{aligned}$$

It follows from Lemma 6 that I doesn't match f. By Lemma 4, if $|I| > n/2 + 1$, f is not a bent function. Note that

$$|I| > n/2 + 1 \Leftrightarrow \Delta_{\mathfrak{R}} < \frac{n/2 - 1}{t+1} \tag{15}$$

and

$$n + 1 - \Delta_{\mathfrak{R}} > t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1) \Leftrightarrow \Delta_{\mathfrak{R}} < \frac{n + t(1-\Gamma_{\mathfrak{R}})}{t+1}. \tag{16}$$

Combining (14), (15) and (16), we obtain item 3.

Case 2.2: $n + 1 - \Delta_{\mathfrak{R}} \leq t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1)$. For this subcase, let

$$\begin{aligned} I = \{&1, 2, \ldots, \Gamma_{\mathfrak{R}} - 1, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}}, \Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} + 1, \ldots, 2\Gamma_{\mathfrak{R}} + \Delta_{\mathfrak{R}} - 2, , \ldots, \\ &(t-1)\Gamma_{\mathfrak{R}} + (t-1)\Delta_{\mathfrak{R}} - (t-2), (t-1)\Gamma_{\mathfrak{R}} + (t-1)\Delta_{\mathfrak{R}} - t + 3, \ldots, \\ &t\Gamma_{\mathfrak{R}} + (t-1)\Delta_{\mathfrak{R}} - t\}, \end{aligned}$$

we have $|I| = t(\Gamma_{\mathfrak{R}} - 1)$. By Lemma 4, if $|I| > n/2 + 1$, then f is not a bent function. Note that

$$|I| > n/2 + 1 \Leftrightarrow \Gamma_{\mathfrak{R}} > \frac{n/2 + 1}{t} + 1 \tag{17}$$

and

$$n + 1 - \Delta_{\mathfrak{R}} \leq t\Gamma_{\mathfrak{R}} + t\Delta_{\mathfrak{R}} - (t-1) \Leftrightarrow \Delta_{\mathfrak{R}} \geq \frac{n + (t-1) - t\Gamma_{\mathfrak{R}}}{t+1}. \tag{18}$$

Combining (17) and (18), we obtain item 4. □

Remark 1 On one hand, for a given RSBF f, if there exists at least one $\mathfrak{R}$ that satisfies one of the conditions in Theorem 3, f cannot be a bent function. When the degree of f is large, such $\mathfrak{R}$ is easy to find. On the other hand, if $\mathfrak{R} = \{R_1, R_2, \ldots, R_k\}$ satisfies one of the conditions in Theorem 3, if $S_i \subset \{0, 1, 2, \ldots, n-1\}$ and contains R_i as a subset for each $1 \le i \le k$, then the corresponding function $f = T_1 + T_2 + \cdots + T_k$ is not a bent function. Thus, we can prove that a large class of homogenous RSBFs is not bent functions.

Example 2 We set $n = 50$ and show that some 50-variable homogenous RSBFs are not bent functions according to Theorem 3.

1. Let f be a 50-variable RSBF with SANF

$$[1, 2, 4, 27, 30, 33] + [1, 3, 6, 17, 27, 35] + [1, 4, 8, 18, 26, 31].$$

Let $\mathfrak{R} = \{R_1, R_2, R_3\}$, where $R_1 = \{1, 4, 27, 33\}$, $R_2 = \{1, 6, 17, 35\}$, $R_3 = \{1, 8, 18, 31\}$. We have $\Gamma_{\mathfrak{R}} = 30$ and $\Delta_{\mathfrak{R}} = 23$. Then item 1 in Theorem 3 is satisfied. Specifically, we set

$$I = \{1, 2, 3, \ldots, 25, 26\}.$$

Note that I doesn't match f and $|I| = 26$. It follows from Lemma 6 that $nl(f) \le 2^{n-1} - 2^{n/2}$. Thus, f is not a bent function. Moreover, for any $S_1, S_2, S_3 \subset \{0, 1, 2, \ldots, 49\}$ with $|S_1| = |S_2| = |S_3|$ and $R_1 \subseteq S_1$, $R_2 \subseteq S_2$, $R_3 \subseteq S_3$, then the corresponding 50-variable RSBF with SANF $T_1 + T_2 + T_3$ is not a bent function.

2. Let f be a 50-variable RSBF with SANF

$$[1, 2, 5, 6, 16, 20] + [1, 2, 4, 8, 17, 21] + [1, 3, 14, 18, 22, 27].$$

Let $\mathfrak{R} = \{R_1, R_2, R_3\}$, where $R_1 = \{1, 5, 16, 20\}$, $R_2 = \{1, 4, 8, 17\}$, $R_3 = \{1, 3, 14, 18\}$. We have $\Gamma_{\mathfrak{R}} = 16$ and $\Delta_{\mathfrak{R}} = 11$. Let $t = 1$, then item 2 in Theorem 3 is satisfied. Specifically, we set

$$I = \{1, 2, \ldots, 14, 15, 28, 29, \ldots, 37, 38\}.$$

Note that I doesn't match f and $|I| = 26$. It follows from Lemma 6 that $nl(f) \le 2^{n-1} - 2^{n/2}$. Thus, f is not a bent function. Moreover, for any $S_1, S_2, S_3 \subset \{0, 1, 2, \ldots, 49\}$ with $|S_1| = |S_2| = |S_3|$ and $R_1 \subseteq S_1$, $R_2 \subseteq S_2$, $R_3 \subseteq S_3$, then the corresponding 50-variable RSBF with SANF $T_1 + T_2 + T_3$ is not a bent function.

3. Let f be a 50-variable RSBF with SANF

$$[1, 3, 5, 12, 18, 22] + [1, 2, 3, 9, 13, 15] + [1, 4, 7, 10, 13, 16].$$

Let $\mathfrak{R} = \{R_1, R_2, R_3\}$, where $R_1 = [1, 5, 12, 18]$, $R_2 = [1, 2, 9, 15]$, $R_3 = [1, 4, 10, 16]$. We have $\Gamma_{\mathfrak{R}} = 14$ and $\Delta_{\mathfrak{R}} = 7$. Let $t = 2$, then item 3 in Theorem 3 is satisfied. Specifically, we set

$$I = \{1, 2, 3, \ldots, 12, 21, 22, 23, \ldots, 32, 41, 42\}.$$

Note that I doesn't match f and $|I| = 26$. It follows from Lemma 6 that $nl(f) \le 2^{n-1} - 2^{n/2}$. Thus, f is not a bent function. Moreover, for any $S_1, S_2, S_3 \subset \{0, 1, 2, \ldots, 49\}$ with $|S_1| = |S_2| = |S_3|$ and $R_1 \subseteq S_1$, $R_2 \subseteq S_2$, $R_3 \subseteq S_3$, then the corresponding 50-variable RSBF with SANF $T_1 + T_2 + T_3$ is not a bent function.

4. Let f be a 50-variable RSBF with SANF

$$[1, 2, 6, 11, 15, 19] + [1, 3, 6, 10, 16, 20] + [1, 2, 4, 7, 13, 16].$$

Let $\mathfrak{R} = \{R_1, R_2, R_3\}$, where $R_1 = [1, 2, 11, 19]$, $R_2 = [1, 6, 16, 20]$, $R_3 = [1, 4, 7, 16]$. We have $\Gamma_{\mathfrak{R}} = 15$ and $\Delta_{\mathfrak{R}} = 10$. Let $t = 2$, then item 4 in Theorem 3 is satisfied. Specifically,

we set

$$I = \{1, 2, 3, \ldots, 13, 25, 26, 27, \ldots, 37\}$$

Note that I doesn't match f and $|I| = 26$. It follows from Lemma 6 that $nl(f) \leq 2^{n-1} - 2^{n/2}$. Thus, f is not a bent function. Moreover, for any $S_1, S_2, S_3 \subset \{0, 1, 2, \ldots, 49\}$ with $|S_1| = |S_2| = |S_3|$ and $R_1 \subseteq S_1$, $R_2 \subseteq S_2$, $R_3 \subseteq S_3$, then the corresponding 50-variable RSBF with SANF $T_1 + T_2 + T_3$ is not a bent function.

We can obtain more nonexistence results of homogeneous RSBF bent functions based on the following lemma.

Lemma 7 [18] *Let f be an n-variable homogeneous RSBF with degree $d \geq 3$. If $\mathcal{M}_f \leq \frac{n}{2}$, then f is not bent.*

For $f(x)$ in (1), let $N^o_{T_i}$ be the number of odd elements in the set S_i and $N^e_{T_i}$ be the number of even elements in the set S_i. Hence, let

$$\begin{aligned} A_f &= \min\{N^o_{T_1}, N^o_{T_2}, \ldots, N^o_{T_k}\}, \\ B_f &= \min\{N^e_{T_1}, N^e_{T_2}, \ldots, N^e_{T_k}\}, \\ C_f &= \max\{N^o_{T_1}, N^o_{T_2}, \ldots, N^o_{T_k}\}. \end{aligned} \tag{19}$$

Theorem 4 *Let n be even and f, A_f, B_f, C_f be defined in (1) and (19). We have the following results.*

1. *If $A_f \geq 2$ and $B_f \geq 2$, then f is not a bent function.*
2. *If $C_f \leq 2$ and $deg(f) \geq \lfloor n/4 \rfloor + 2$, then f is not a bent function.*

Proof 1. Let's consider the set

$$I = \{0, 1, 2, 4, 6, 7, \ldots, n-4, n-2\}.$$

Note that $|I| = n/2 + 1$. If $A_f \geq 2$ and $B_f \geq 2$, it is easy to see that I doesn't match f. By Lemma 4, f is not a bent function.

2. Let $C_f \leq 2$. Assume that f is a bent function with degree $d \geq \lfloor n/4 \rfloor + 2$. Next we construct a set $I \subseteq \{0, 1, 2, \ldots, n-1\}$, which matches f and has size $n/2 + 1$.

Let Z be the integer ring. Define a vector in Z^{2n-2} as

$$\mathfrak{V} = (0, 1, 2, 3, \ldots, n-1, 0, 1, 2, \ldots, n-3).$$

For $1 \leq s \leq n-1$ and $0 \leq r \leq n-1$, let

$$V_{r,s} = \{r, (r+1) \bmod n, (r+2) \bmod n, \ldots, (r+s-1) \bmod n\},$$

that is, a set of s consecutive integers in $\mathfrak{V}$ started with r.

The discussion is divided into two cases based on the value of $N^o_{T_i}$ where T_i generates the largest distance of f.

Case 1: there exists T_i in the SANF of f such that $N^o_{T_i} \leq 1$ and T_i generates the largest distance of f. Denote by

$$S^l_i = \{l, (t^{(i)}_1 + l) \bmod n, (t^{(i)}_2 + l) \bmod n, \ldots, (t^{(i)}_{d-1} + l) \bmod n\}$$

for $0 \leq l \leq n-1$. We set

$$I = \{0, 1, 2, 4, 6, 8, \ldots, n-4, n-2\},$$

then $|I| = n/2 + 1$ and I matches a term in $G_n(x_0 x_{t_1^{(i)}} \cdots x_{t_{d-1}^{(i)}})$, that is, there exists l such that $S_i^l \subseteq I$. According to Lemma 7, we have

$$\mathcal{M}_f = \mathcal{D}_{S_i^l} \geq 2/n + 1.$$

It is easy to see that $|V_{r,\mathcal{D}_{S_i^l}} \cap I| \geq \lceil n/4 \rceil$ holds for all r. Thus, we have

$$d = |S_i^l| \leq n/2 + 1 - \lceil n/4 \rceil = \lfloor n/4 \rfloor + 1,$$

a contradiction.

Case 2: there is no T_i in f such that $N_{T_i}^o \leq 1$ and T_i generates the largest distance of f. This case implies that, if T_i generates the largest distance of f, then $N_{T_i}^o = 2$. Assume that the term $x_{2i_1} x_{2i_2} \cdots x_{2i_{d-2}} x_{2s_1+1} x_{2s_2+1}$ in f generates the largest distance of f, where $2i_1 < 2i_2 < \cdots < 2i_{d-2} \leq 2n - 4$. We set

$$I = \{0, 2, 4, \ldots, 2n - 6, 2n - 4, 2s_1 + 1, 2s_2 + 1\}.$$

One can see that I matches $x_{2i_1} x_{2i_2} \cdots x_{2i_{d-2}} x_{2s_1+1} x_{2s_2+1}$. Denote by

$$U = \{2i_1, 2i_2, \ldots, 2i_{d-2}, 2s_1 + 1, 2s_2 + 1\}.$$

By Lemma 7, we have

$$\mathcal{M}_f = \mathcal{D}_U \geq 2/n + 1.$$

It is easy to see that $|\mathcal{D}_U \cap I| \geq \lceil n/4 \rceil$, which implies that

$$d = |S_i^l| \leq n/2 + 1 - \lceil n/4 \rceil = \lfloor n/4 \rfloor + 1,$$

a contradiction. □

3.3 Nonexistence results of cubic homogeneous rotation symmetric bent functions

In this subsection, we investigate the bentness of cubic homogeneous rotation symmetric functions. Let

$$f_n(x) = [0, r_1, s_1] + [0, r_2, s_2] + \cdots + [0, r_k, s_k] \tag{20}$$

be an n-variable cubic RSBF, which can be represented by the set

$$M = \{(r_1, s_1), (r_2, s_2), \ldots, (r_k, s_k)\}.$$

Assume that $s_i > r_i$ for $1 \leq i \leq k$. We have

$$\Delta_{1^n} f_n(x) = \sum_{i=1}^{k} [0, r_i] + \sum_{s=1}^{k} [0, s_i] + \sum_{i=1}^{k} [0, s_i - r_i]. \tag{21}$$

Note that $[0, i] = [0, j]$ if $i + j = n$. When we eliminate the terms that occur an even number of times in (21), $\Delta_{1^n} f_n(x)$ is reduced to $\Delta_{1^n} f_n(x) = \sum_{i \in I_{M,n}} [0, i]$, where $I_{M,n}$ is determined by M and n. If f_n is a bent function, then $\sum_{x \in \mathbb{F}_2^n} (-1)^{\Delta_{1^n} f_n(x)} = 0$, which implies that $\Delta_{1^n} f_n(x)$ is balanced on $\mathbb{F}_2^n$. Next, we investigate the balancedness of $\Delta_{1^n} f(x)$ by converting it to a quadratic idempotent over $\mathbb{F}_{2^n}$.

Let g be an n-variable RSBF, define a function g' over $\mathbb{F}_{2^n}$ as $g'(z) = g(z, z^2, \ldots, z^{2^{n-1}})$. It is shown in [3] that g' is an idempotent over $\mathbb{F}_{2^n}$ and g' has the same degree and nonlinearity with g. Let $Q_n(x) = \sum_{i\in\{i_1,i_2,\ldots,i_k\}}[0, i]$ be a quadratic RSBF, then

$$Q'_n(z) = \sum_{i\in\{i_1,i_2,\ldots,i_k\}} \mathrm{Tr}_1^n(x^{2^i+1}). \tag{22}$$

Hence, Q_n is balanced if and only if Q'_n is balanced. In [5], some profound results on the balancedness of Q' are presented.

Lemma 8 [5] *Let Q'_n be defined in (22). We have the following results.*

1. *If $v_2(n) \le \{v_2(i_1), v_2(i_2), \ldots, v_2(i_k)\}$, then Q'_n is balanced if and only if k is odd.*
2. *There is a finite set $S(Q'_n)$ of positive integers such that Q'_n is balanced if and only if*

$$n \equiv 2^{d-1} \bmod 2^d \ \textit{for some}\ d \in S(Q'_n).$$

If $f_n(x)$ in (20) has an even number of terms, that is, k is even, then $\Delta_{1^n} f(x)$ in (21) also have an even number of terms. We obtain the following result immediately.

Theorem 5 *Let $f_n(x)$ be defined in (20) and k be even. Assume that $v_2(r_i), v_2(s_i) \ge v_2(n)$ for all $1 \le i \le k$. Then f is not bent.*

Theorem 6 *Given positive integer pairs $P = \{(r_1, s_1), (r_2, s_2), \ldots, (r_k, s_k)\}$ with $s_i > r_i$ for $1 \le i \le k$, there exists an integer e_0 such that the cubic n-variable homogeneous RSBF*

$$f_n(x) = [0, r_1, s_1] + [0, r_2, s_2] + \cdots + [0, r_k, s_k]$$

is not bent for any $n \equiv 0 \bmod 2^{e_0}$.

Proof Let $m = \max\{s_1, s_2, s_3, \ldots, s_k\}$ and

$$V = \{r_1, r_2, \ldots, r_k\} \bigcup \{s_1, s_2, \ldots, s_k\} \bigcup \{s_1 - r_1, s_2 - r_2, \ldots, s_k - r_k\}.$$

Assume that $n > 2m$. Then $i + j = n$ cannot occur for any $i, j \in V$. Thus, $\Delta_{1^n} f_n(x)$ in (21) is independent of n and is generated by P, which can be represented by $\Delta_{1^n} f_n(x) = \sum_{i\in I_P}[0, i]$, where $I_P \subseteq V$. By Lemma 8, there exists S such that $\Delta_{1^n} f_n(x)$ is balanced if and only if $n \equiv 2^{d-1} \bmod 2^d$ for some $d \in S$. In fact, S is only determined by P. Let n_1 be the maximal element in set S. If $v_2(n) \ge n_1$, $n \equiv 0 \bmod 2^d$ for any $d \in S$. Thus, $\Delta_{1^n} f_n(x)$ cannot be balanced.

Let n_2 be the integer such that $2^{n_2-1} \le m < 2^{n_2}$ and set $e_0 = \max\{n_1, n_2 + 1\}$. Then $2^{e_0} > 2m$. Hence, $\Delta_{1^n} f_n(x)$ cannot be balanced if $n \equiv 0 \bmod 2^{e_0}$. Thus, f is not bent. □

Example 3 Let $n \ge 10$ and f be an n-variable cubic homogeneous rotation symmetric function with SANF $f = [0, 1, 2] + [0, 1, 3] + [0, 1, 4] + [0, 1, 5]$. We have

$$\Delta_{1^n} f_n(x) = [0, 1] + [0, 5].$$

According to [5], the function $[0, 1] + [0, 5]$ cannot be balanced when $n \equiv 0 \bmod 4$. Thus, we set $e_0 = 4$ in Theorem 6, f cannot be bent when $n \equiv 0 \bmod 2^4$.

The following nonexistence results on cubic homogeneous rotation symmetric bent functions can be obtained easily by Theorem 2.

Corollary 2 *Let $n = 2p_1^{n_1} p_2^{n_2} p_3^{n_3} \cdots p_t^{n_t}$ and $p_i > 2^{2p_1^{n_1}\cdots p_{i-1}^{n_{i-1}}}$ for any $2 \le i \le t$, there are no n-variable cubic homogeneous rotation symmetric bent functions.*

Table 1 Known results on the nonexistence of homogeneous rotation symmetric bent functions

No	Conditions			Refs
	n	d	T_i	
1	-	$n/2$	-	[31]
2	-	≥ 3	$k = 1$	[25]
3	-	≥ 3	$\mathcal{M}_f \leq \frac{n/2-1}{[n/d]}$	[25]
4	-	≥ 3	$\mathcal{M}_f \leq \frac{n}{2}$	[18]
5	-	-	$S_1 = \{1, \ldots, d\}$, $\exists i \geq 2$ such that $S_i \neq \{1, \ldots, l, t^i_{d-1} - d + l + 1, \ldots, t^i_{d-1}\}$	[33]
6	$2p$	> 2	$r = 0$ or s is odd	[8]
7	$2p$	≥ 4	$2 \leq N^o_{T_i} \leq d - 2$ holds for all i with $\lvert G_n(T_i) = n \rvert$	[8]
8	$p_1 > 2^{2^{n_0}}$, for $i \geq 2$ $p_i > 2^{2^{n_0} p_1^{n_1} \cdots p_{i-1}^{n_{i-1}}}$	-	if $n_0 = 1$, N_1 is even if $n_0 = 2$, N_1 is odd or $N_2 + N_3$ is even if $n_0 \geq 3$, N_1 is odd or $N_2 + N_3$ is odd	this paper
9	$n_0 = 1$, for $i \geq 2$ $p_i > 2^{2^{n_0} p_1^{n_1} \cdots p_{i-1}^{n_{i-1}}}$	d is odd	-	this paper
10	-	-	$n - \Gamma_{\mathfrak{R}} < \Delta_{\mathfrak{R}} < n/2 - 1$	this paper

Table 1 continued

No	Conditions			Refs
	n	d	T_i	
11	-	-	$\exists t \geq 1$ such that $\frac{n+t-(t+1)\Gamma_{\mathfrak{R}}}{t+1} < \Delta_{\mathfrak{R}}$ $\Delta_{\mathfrak{R}} < \min\{\frac{n+(1-\Gamma_{\mathfrak{R}})(t+1)}{t}, \frac{n/2}{t+1}\}$	this paper
12	-	-	$\exists t \geq 1$ such that $\Delta_{\mathfrak{R}} < \frac{n/2-1}{t+1}$, $\Delta_{\mathfrak{R}} < \frac{n+t(1-\Gamma_{\mathfrak{R}})}{t+1}$ $\frac{n+(t+1)(1-\Gamma_{\mathfrak{R}})}{t} \leq \Delta_{\mathfrak{R}}$	this paper
13	-	-	$\exists t \geq 1$ such that $\Gamma_{\mathfrak{R}} \geq \frac{n/2+1}{t} + 1$, $\Delta_{\mathfrak{R}} < \frac{n+t-1-t\Gamma_{\mathfrak{R}}}{t}$ $\frac{n+t-1-t\Gamma_{\mathfrak{R}}}{t+1} \leq \Delta_{\mathfrak{R}}$ $\frac{n+(t+1)(1-\Gamma_{\mathfrak{R}})}{t} \leq \Delta_{\mathfrak{R}}$	this paper
14	-	-	$A_f, B_f \geq 2$	this paper
15	-	$\geq \lfloor n/4 \rfloor + 2$	$C_f \leq 2$	this paper
16	$n \equiv 0 \bmod 2^{e_0}$	3	-	this paper
17	$n_0 = 1$ $p_i > 2^{2p_1^{n_1} \cdots p_{i-1}^{n_{i-1}}}$	3	-	this paper

4 Conclusion and remarks

In this paper, we study the conjecture that there are no homogeneous rotation symmetric bent functions with degree larger than 2 from some different viewpoints and obtain new nonexistence results on homogeneous rotation symmetric bent functions. How to solve the conjecture is still an extremely difficult problem. In the future, we will continue to study this conjecture by developing some new methods.

To advance future research, we list the main known results on the conjecture in Table 1. For consistency, let $n = 2^{n_0} p_1^{n_1} \cdots p_t^{n_t}$ and f be an n-variable homogeneous RSBF as defined in (1). Let $r = |\{1 \le i \le k : |G_n(T_i)| = n\}|$ and $s = |\{1 \le i \le k : |G_n(T_i)| < n\}|$. Let S_i, $\mathcal{M}_f$, $\Delta_{\mathfrak{R}}$, $\Gamma_{\mathfrak{R}}$, N_i, A_f, B_f, C_f, $N_{T_i}^o$ be as defined in this paper. The conditions on n, d and T_i such that f cannot be a bent function are summarized in Table 1.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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